

Submission PHYS1-MID-SUB02

Question PHYS1-MID-Q01

work / setup:

1. $v = v_{\text{initial}} + at = 4 + 3(5) = 19 \text{ m/s}$

crossed-out arithmetic: [smudged]

FINAL (maybe): $v = 19 \text{ m/s}$; displacement = 57.5 m plus 1

Question PHYS1-MID-Q02

work / setup:

1. $N = mg = 58.8 \text{ N}$; friction = $\mu N = 11.8 \text{ N}$

2. $\Sigma F = 17 - 11.8 = 5.2 \text{ N}$

crossed-out arithmetic: [smudged]

FINAL (maybe): $a = 0.87 \text{ m/s}^2$; friction opposes motion plus 1

Question PHYS1-MID-Q03

work / setup:

1. $mgh = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{(2gh)} = \sqrt{(2 \cdot 9.8 \cdot 8)}$

crossed-out arithmetic: [smudged]

FINAL (maybe): $v = 12.52 \text{ m/s}$ plus 1

Question PHYS1-MID-Q04

work / setup:

1. $p_{\text{total, before}} = p_{\text{total, after}}$

crossed-out arithmetic: [smudged]

FINAL (maybe): total momentum before equals after when external impulse is negligible plus 1

Question PHYS1-MID-Q05

work / setup:

1. Impulse is area under $F(t)$: $J = F\Delta t = 6(0.5) = 3 \text{ N}\cdot\text{s}$.

crossed-out arithmetic: [smudged]

FINAL (maybe): impulse = 3 N·s; momentum changes by 3 kg·m/s plus 1

Question PHYS1-MID-Q06

work / setup:

1. $v = v_{\text{initial}} + at = 7 + 3(2) = 13 \text{ m/s}$

crossed-out arithmetic: [smudged]

FINAL (maybe): $v = 13 \text{ m/s}$; displacement = 20 m plus 1

Question PHYS1-MID-Q07

work / setup:

1. $N = mg = 19.6 \text{ N}$; friction = $\mu N = 3.9 \text{ N}$

2. $\Sigma F = 16 - 3.9 = 12.1 \text{ N}$

crossed-out arithmetic: [smudged]

FINAL (maybe): $a=6.04 \text{ m/s}^2$; friction opposes motion plus 1

Question PHYS1-MID-Q08

work / setup:

1. $mgh = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2gh} = \sqrt{2 \cdot 9.8 \cdot 9}$

crossed-out arithmetic: [smudged]

FINAL (maybe): $v=13.28 \text{ m/s}$ plus 1